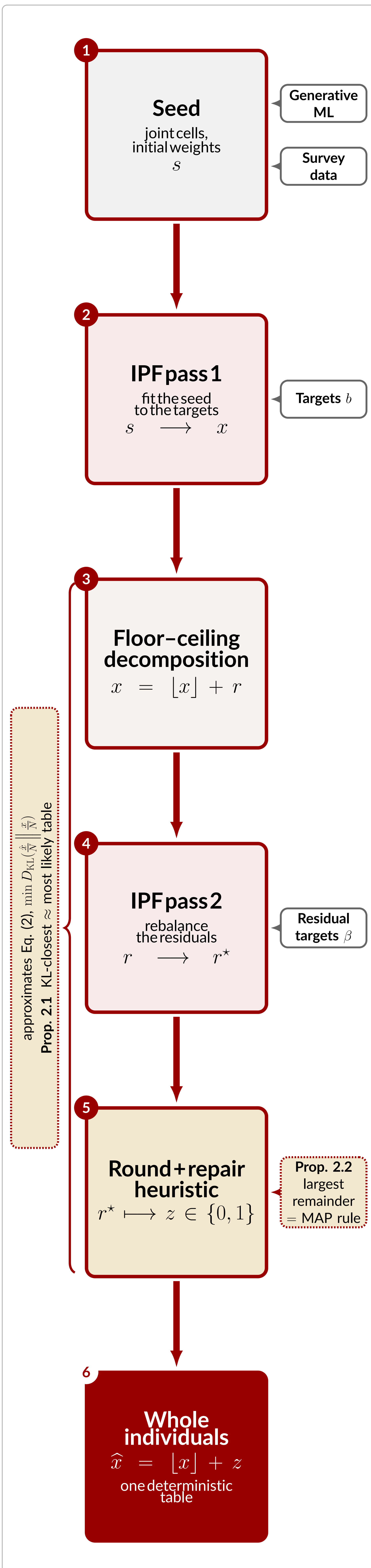




Introduction

- ▶ Agent-based transport models need whole individuals. The problem grows as population tables get sparser, because a large share of the population then sits in cells with fractional counts.
- ▶ Uninformed sampling creates two problems: (i) the margins do not hold, and (ii) the sampling error is large (Lovelace and Ballas, 2013).
- ▶ In forecasting, a scenario difference should reflect the policy, not the random seed.
- ▶ **Idea**: among all integer tables that meet the margins, choose the one closest to the IPF table in KL divergence.
- ▶ Hence, the method extends the KL projection of IPF (Deming and Stephan, 1940) to integer allocation, close to the KL optimum.

Methodology

- ▶ **The problem**, Eq. (2):

$$\min_{\hat{x} \in \mathbb{Z}_{\geq 0}} D_{\text{KL}}\left(\frac{\hat{x}}{N} \parallel \frac{x}{N}\right)$$

$$\text{s.t. } \sum_{k \in \mathcal{K}(s)} \hat{x}_k = b_s$$

- ▶ **Why hard**: a nonlinear integer program with one variable per cell; with overlapping margins it is not guaranteed to be totally unimodular.
- ▶ **Solution approach**: in the spirit of KL divergence and almost optimal, within 3.6% of the KL optimum.

- ▶ **Two passes**: IPF fits the seed, then rebalances the residuals.
- ▶ **Floor-ceiling**: $\hat{x} = \lfloor x \rfloor + z, z \in \{0, 1\}$.
- ▶ **Round + repair**: largest remainder, then ± 1 swaps.

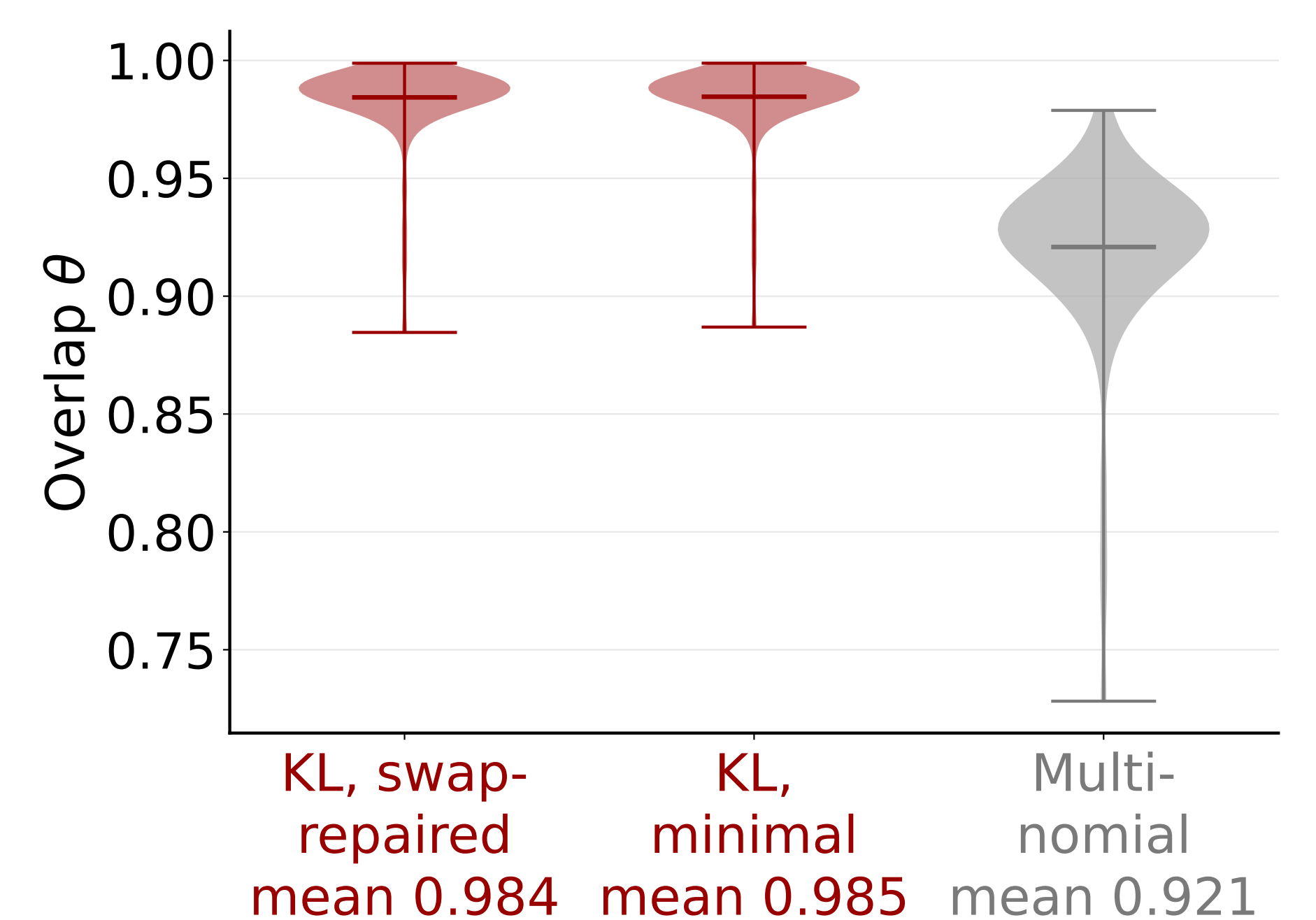
Application and implications

- ▶ **Denmark**: 98 municipalities, 635,000 joint cells, 5.9 million persons.
- ▶ **KL allocation, no noise**: **98.4%** of the mass in place, against **92.1%** for multinomial sampling, with zero run-to-run variance (Fig. 5).
- ▶ **Margins**: the controlling targets hold exactly; the secondary margins give way, by 1.4% (Table 3).
- ▶ **Scalable**: 397 million cells in 16 minutes; the controlled table keeps $\theta_B = 0.984$ (Table 4).

Implications

- ▶ No algorithmic sampling noise: scenario differences reflect the policy.
- ▶ 292,000 fewer persons misplaced than with multinomial sampling (53,000 against 345,000).

Overlap with the fractional table (Fig. 5)



$$\theta = \sum_k \min(p_k, q_k), 98 \text{ municipalities; multinomial: median of 200 draws}$$

Benchmark (Table 3)

Approaches without a repair mechanism sacrifice the target margins

Method	θ	std	Target	Second.	Rare
KL, swap-repaired	0.984	0	0.00	1.39	0.92
Hamilton	0.985	0	0.29	1.98	0.87
TRS	0.977	0.0002	0.28	1.75	1.05
Multinomial	0.921	0.0016	1.70	8.40	1.00

Margin errors in % of population; rare = mass kept in cells with $x_k < 1$

Scaling (Table 4)

As the table gets sparser, the gap between the fractional IPF table and any integer table grows

k	Cells	Floor	full-table θ	
			KL	Multin.
0	0.63M	0.95	0.984	0.921
1	3.2M	0.87	0.949	0.850
2	15.9M	0.73	0.880	0.741
3	79.4M	0.53	0.759	0.588
4	397M	0.31	0.592	0.413

k added attributes. Controlled table: KL 0.984, multinomial 0.921 at every k .